

Question Set 2

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here "Reflecting questions" (RQ). The questions more similar to exam/test questions will be labelled ETQ.

In class questions:

- 1) Look at the definition of T_g in the notes or the recommended textbook. What will cause increase or decrease of T_g ? (Try to think of the molecular level, and think of the molecular dynamics and mobility that T_g is related to: try to think of the physics involved **first**, and, **then**, how chemical changes to structure will affect these) Discuss briefly and be prepared to share your answers.

rotational mobility increase (facilitate) $\uparrow$ $T_g \downarrow$

rotational mobility hinder $\downarrow$ $T_g \uparrow$

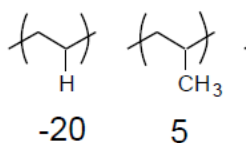
RQ: At molecular level, T_g is determined primarily by rotational mobility of the main chain (and to a much lesser extent, vibrational mobility, which for the purpose of connecting to the chemical structure we will ignore as it is not as important as rotational mobility). Whatever facilitates rotational mobility will decrease T_g (as less energy will be needed for the transition to occur). Whatever hinders rotational mobility will increase T_g (as more energy is needed for the transition to occur). SO: groups attached to the main chain will tend to decrease the rotational mobility of the main chain (increase T_g). However, as we are not looking at 1 isolated chain alone, but we're observing a material (with MANY molecules), we need to understand that these groups will have an impact on how close the chains are to each other: how much entanglement they cause. Increased entanglement will hinder the rotational mobility of the main chain, increasing T_g . The bigger the group, the more free volume between main chains it will produce (because the chains cannot come closer together), THEREFORE, there is an opposing effect, which results in bigger groups producing more free volume, facilitating rotational mobility and decreasing T_g . Your task is to see which one wins (but the values of T_g will be given).

the group bigger more free volume between main chain it will produce

The other effects are: all attracting secondary interactions bring chains together: hindering rotational mobility, therefore, increasing T_g .

facilitating rotational mobility $T_g \downarrow$ decrease

- 2) Explain the T_g changes shown in the following picture (data in $^{\circ}\text{C}$):



ETQ: In PP, the methyl group hinders rotational mobility of the polymer main chain (there is no side group in PE), this increases T_g as shown by the values.

A exam full answer for full marks would include (SPRE):

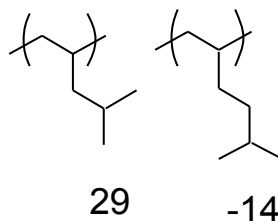
S-An explanation of what's important at molecular level (see question 1, part in bold in red)

P-A brief description of what matters here (see the structure of the polymers: the only difference is one has a group CH_3 , the other one doesn't, the one with the group has higher T_g than the other one, therefore, this is simply the group directly affecting the rotational mobility (see Question 1, part in bold in blue)

R-Answer as in above (see this question highlighted in yellow)

E-answer as in above (see this question, the little part not highlighted in yellow).

3) Explain the T_g changes shown in the following picture (data in °C):



ETQ: Both polymers have groups attached to the main chain. The first polymer has a branching unit closer to the main chain: this hinders the rotational mobility of the main chain (increases T_g), the second polymer has a branching unit slightly further away from the main chain. This pushes other molecules away, creating free volume, which facilitates the rotational mobility of the main chain (decreasing the T_g). Note that branching units create bigger groups.

A exam full answer for full marks would include (SPRE):

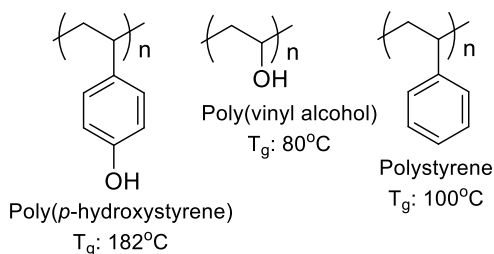
S-An explanation of what's important at molecular level (see question 1, part in bold in red)

P-A brief description of what matters here (see the structure of the polymers: the only difference is that the first polymer has a branching unit closer to the main chain, while the second is further away. This makes the impact on the first polymer more pronounced on the main chain of the polymer itself (see Question 1, part in bold in blue), while on the second polymer the impact of the branching unit is more on other molecules creating free volume. (see Question 1, part in bold in green)

R/E-Answer as in above (see this question highlighted in yellow)

4) Explain the T_g changes shown in the following picture (data in °C):

a.



ETQ: We have three comparisons possible (in an exam, you will be asked separately to consider only pairs, so it will be obvious which pair you need to compare). Here the comparisons that are meaningful are PHS-PVA and PHS-PS

PHS: Poly(p-hydroxystyrene); PVA: Poly(vinyl alcohol); PS (Polystyrene)

PHS-PVA: The difference between the two polymers is that PHS has a benzene ring, while PVA has not. The rest is the same. This means, PHS has an additional secondary interaction (attracting forces) between side chains and molecules. This is pi-pi (π - π) stacking. Therefore, as there are more attracting forces between main chains in PHS compared to PVA, this increases entanglement and decreases rotational mobility, increasing T_g (almost 100°C increase).

A exam full answer for full marks would include (SPRE):

S-An explanation of what's important at molecular level (see question 1, part in bold in red, particularly the one underlined!!!)

P-A brief description of what matters here (see the structure of the polymers: the only difference the benzene ring)

R/E-Answer as in above (see this question highlighted in yellow)

PHS-PS: The difference between the two polymers is that PHS has an OH group, while PS has not. The rest is the same. This means, PHS has an additional secondary interaction (attracting forces) between side chains and molecules. This is H-bonding (strong) and polarity (medium, as the electronegativity difference between C-O is large). Therefore, as there are more attracting forces between main chains in PHS compared to PS, this increases entanglement and decreases rotational mobility, increasing T_g (almost 80°C increase).

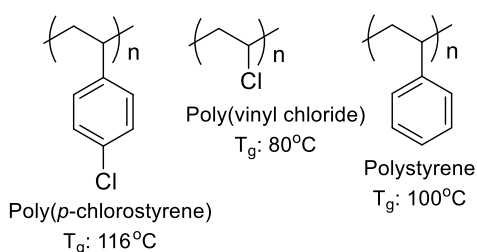
A exam full answer for full marks would include (SPRE):

S-An explanation of what's important at molecular level (see question 1, part in bold in red, particularly the one underlined!!!)

P-A brief description of what matters here (see the structure of the polymers: the only difference is the OH group)

R/E-Answer as in above (see this question highlighted in yellow)

b.



ETQ: We have three comparisons possible (in an exam, you will be asked separately to consider only pairs, so it will be obvious which pair you need to compare). Here the comparisons that are meaningful are PCS-PVC and PCS-PS

PCS: Poly(p-chlorostyrene); PVC: Poly(vinyl chloride); PS (Polystyrene)

PCS-PVC: The difference between the two polymers is that PCS has a benzene ring, while PVC has not. The rest is the same. This means, PCS has an additional secondary interaction (attracting forces) between side chains and molecules. This is pi-pi (π - π) stacking. Therefore, as there are more attracting forces between main chains in PCS compared to PVC, this increases entanglement and decreases rotational mobility, increasing T_g (almost 100°C increase).

A exam full answer for full marks would include (SPRE):

S-An explanation of what's important at molecular level (see question 1, part in bold in red, particularly the one underlined!!!)

P-A brief description of what matters here (see the structure of the polymers: the only difference the benzene ring)

R/E-Answer as in above (see this question highlighted in yellow)

PCS-PS: The difference between the two polymers is that PCS has a Cl, while PS has not. The rest is the same. This means, PCS has an additional secondary interaction (attracting forces) between side chains and molecules. This some level of polarity (weak, as the electronegativity difference C-Cl is not too large). Therefore, as there are some more attracting forces between main chains in PCS compared to PS, this increases entanglement and decreases rotational mobility, increasing T_g (the effect is weak, therefore, only some 15°C increase).

A exam full answer for full marks would include (SPRE):

S-An explanation of what's important at molecular level (see question 1, part in bold in red, particularly the one underlined!!!)

P-A brief description of what matters here (see the structure of the polymers: the only difference is the Cl)

R/E-Answer as in above (see this question highlighted in yellow)